Lecture 8. Limit Theorems

Markov and Chebyshev, weak and strong laws, central limit theorem, Chernoff bounds, Jensen

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8.1 Markov and Chebyshev inequalities; the weak law

Motivation. We often know only a mean or variance. Markov's and Chebyshev's inequalities give bounds on tail probabilities from these alone, and Chebyshev immediately shows that sample averages concentrate around the mean.

Proposition 8.1 (Markov's inequality)

If $X \geq 0$ and $a > 0$, then $P(X \geq a) \leq E[X]/a$.

Proof. $a\mathbb{1}\{X \geq a\} \leq X$; take expectations. ∎

Proposition 8.2 (Chebyshev's inequality)

If X has mean μ and variance σ^2, then $P(|X - \mu| \geq k) \leq \sigma^2/k^2$ for $k > 0$.

Proof. Apply Markov's inequality to $(X - \mu)^2$ and $a = k^2$. ∎

These bounds use only the mean (and variance), so they are often crude, but they hold for every distribution.

Example 8.1

(a) Weekly output has mean 50. Then $P(X \geq 75) \leq 50/75 = 2/3$. If also $\mathrm{Var}(X) = 25$, $P(40 < X < 60) \geq 1 - 25/100 = 3/4$.
(b) For $X \sim U(0, 10)$, Chebyshev gives $P(|X - 5| > 4) \leq (25/3)/16 \approx 0.52$, whereas the true value is 0.2.
(c) If $\mathrm{Var}(X) = 0$, then $P(|X - \mu| > 1/n) = 0$ for all n, and by continuity $P(X = \mu) = 1$.

Theorem 8.3 (Weak law of large numbers)

If $X_1, X_2, \ldots$ are i.i.d. with mean μ, then for every $\epsilon > 0$, $P(|\bar{X}_n - \mu| \geq \epsilon) \to 0$ as $n \to \infty$.

Proof. Assuming finite variance σ^2: $E[\bar{X}_n] = \mu$, $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$, and Chebyshev gives the bound $\sigma^2/(n\epsilon^2) \to 0$. ∎

8.2 The central limit theorem

Motivation. The law of large numbers says averages converge; the central limit theorem says how fluctuations around the mean look: approximately normal, whatever the original distribution. This is why the normal distribution is everywhere in statistics.

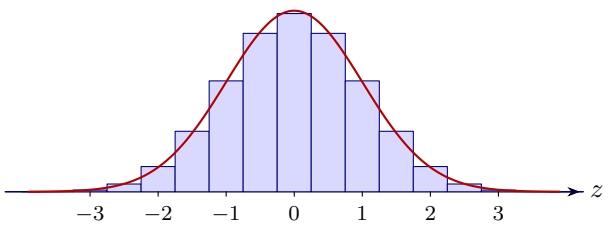


Figure. *The distribution of the standardized sum* $(S_{16} - 8)/2$ *for* $S_{16} \sim \mathrm{Bin}(16, \frac{1}{2})$ *(bars, scaled to unit area) against the standard normal density. Already for* $n = 16$ *the CLT approximation is close.*

Theorem 8.4 (Central limit theorem)

If $X_1, X_2, \dots$ are i.i.d. with mean μ and variance $\sigma^2 \in (0, \infty)$, then for every a,

$$P\Big(\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \le a\Big) \to \Phi(a) \qquad (n \to \infty).$$

The key tool is: if $M_{Z_n}(t) \to M_Z(t)$ for all t, then $F_{Z_n}(a) \to F_Z(a)$ at continuity points of F_Z (quoted).

Proof sketch. Take $\mu = 0$, $\sigma = 1$, and assume the mgf M of X_i exists near 0. Let $L(t) = \log M(t)$; then $L(0) = 0$, $L'(0) = \mu = 0$, $L''(0) = E[X^2] - \mu^2 = 1$. The mgf of $Z_n = \sum X_i/\sqrt{n}$ is $M(t/\sqrt{n})^n$, and by L'Hôpital twice (in $x = 1/\sqrt{n}$),

$$nL\Big(\frac{t}{\sqrt{n}}\Big) = \frac{L(tx)}{x^2} \to \frac{t^2 L''(0)}{2} = \frac{t^2}{2}.$$

So $M_{Z_n}(t) \to e^{t^2/2}$, the $N(0,1)$ mgf. (Without an mgf the same argument works with characteristic functions.) ∎

The theorem explains why measurement errors, heights, and sums of many small effects are approximately normal. It also holds for independent, non-identically distributed X_i that are uniformly bounded with $\sum \mathrm{Var}(X_i) = \infty$: then $(\sum(X_i - \mu_i))/\sqrt{\sum \sigma_i^2}$ is approximately $N(0,1)$.

Example 8.2 (How many measurements?)

An astronomer averages n independent measurements with variance 4. To be 95% sure that the average is within 0.5 of the true value, the CLT requires $P(|Z| \le 0.5\sqrt{n}/2) \approx 0.95$, i.e. $0.25\sqrt{n} \ge 1.96$, $n \ge 62$. Chebyshev's inequality would demand $4/(0.25n) \le 0.05$, i.e. $n \ge 320$: guaranteed for every distribution, but far more conservative.

Example 8.3

(a) The sum S of 10 fair dice has mean 35 and variance $350/12$; with the continuity correction, $P(30 \le S \le 40) \approx 2\Phi(5.5/5.40) - 1 \approx 0.69$.
(b) The sum of 10 independent $U(0,1)$ variables has mean 5 and variance $10/12$, so $P(\sum X_i > 6) \approx 1 - \Phi(1.10) \approx 0.137$.
(c) Exam grading times are i.i.d. with mean 20 and standard deviation 4 minutes. The probability that the first 25 exams are graded within 450 minutes is approximately $\Phi\big((450 - 500)/20\big) = \Phi(-2.5) \approx 0.006$.

8.3 The strong law of large numbers

Motivation. The weak law concerns each fixed n; the strong law says that the sequence of averages itself converges with probability one, which justifies the frequency interpretation of probability.

Theorem 8.5 (Strong law of large numbers)

If $X_1, X_2, \dots$ are i.i.d. with finite mean μ, then $P\big(\lim_{n\to\infty} \bar{X}_n = \mu\big) = 1$.

Proof. Under the extra assumption $K = E[X_i^4] < \infty$, with $\mu = 0$. Expanding $E[S_n^4]$, the terms with some index appearing exactly once vanish, leaving $nE[X^4] + 6\binom{n}{2}E[X_i^2 X_j^2] \le nK + 3n(n-1)K$

(as $E[X^2]^2 \le E[X^4]$). Hence $E[\sum_n S_n^4/n^4] \le \sum_n (K/n^3 + 3K/n^2) < \infty$, so $\sum_n S_n^4/n^4 < \infty$ with probability 1, which forces $S_n^4/n^4 \to 0$, i.e. $\bar{X}_n \to 0$. ■

The weak law says that for each large n, $\bar{X}_n$ is *probably* close to μ; the strong law says that, with probability 1, the whole sequence eventually stays close to μ. Applied to $X_i = \mathbb{1}\{E \text{ occurs on trial } i\}$, it justifies the frequency interpretation: the long-run proportion of trials on which E occurs equals $P(E)$ with probability 1.

8.4 Other inequalities

Motivation. Sharper or one-sided bounds are often needed: Chernoff bounds give exponential tails from the mgf, and Jensen's inequality compares $E[g(X)]$ with $g(E[X])$ for convex g.

Proposition 8.6 (One-sided Chebyshev)

If $E[X] = 0$ and $\mathrm{Var}(X) = \sigma^2$, then $P(X \ge a) \le \sigma^2/(\sigma^2 + a^2)$ for $a > 0$.

Proof. For $b > 0$, $P(X \ge a) \le P((X+b)^2 \ge (a+b)^2) \le (\sigma^2 + b^2)/(a+b)^2$ by Markov; choose $b = \sigma^2/a$. ■

Proposition 8.7 (Chernoff bounds)

If $M(t) = E[e^{tX}]$, then $P(X \ge a) \le e^{-ta}M(t)$ for all $t > 0$ and $P(X \le a) \le e^{-ta}M(t)$ for all $t < 0$.

Proof. For $t > 0$, $P(X \ge a) = P(e^{tX} \ge e^{ta}) \le E[e^{tX}]e^{-ta}$ by Markov. ■

One then minimizes over t.

Example 8.4

(a) Standard normal: $e^{-ta+t^2/2}$ is minimized at $t = a$, so $P(Z \ge a) \le e^{-a^2/2}$ for $a > 0$.
(b) Poisson(λ) and $i > \lambda$: minimizing $e^{\lambda(e^t-1)-ti}$ at $e^t = i/\lambda$ gives $P(X \ge i) \le e^{-\lambda}(e\lambda)^i/i^i$.
(c) A gambler wins or loses 1 with probability $\frac{1}{2}$ each play. After n plays, S_n has $M(t) = \cosh(t)^n \le e^{nt^2/2}$, so $P(S_n \ge a) \le e^{-a^2/(2n)}$ (take $t = a/n$); e.g. $P(S_{10} \ge 6) \le e^{-1.8} \approx 0.165$, while the exact value is 0.0547.

Proposition 8.8 (Jensen's inequality)

If f is convex, then $E[f(X)] \ge f(E[X])$ whenever the expectations exist.

Proof. With $\mu = E[X]$, convexity gives $f(x) \ge f(\mu) + f'(\mu)(x - \mu)$ (for twice differentiable f, by Taylor's theorem with $f'' \ge 0$). Take expectations. ■

For example, an investor choosing between a sure return μ and a random return X with mean μ, with concave utility u, prefers the sure amount since $E[u(X)] \le u(\mu)$.

8.5 The Poisson approximation, quantified

Let $X = \sum_{i=1}^n X_i$ with independent $X_i \sim$ Bernoulli(p_i), and $Y \sim$ Poisson(λ) with $\lambda = \sum p_i$. Coupling each X_i with a Poisson(p_i) variable Y_i so that they agree with probability at least $1 - p_i^2$,

one obtains, for every set A of integers,

$$|P(X \in A) - P(Y \in A)| \le \sum_{i=1}^{n} p_i^2.$$

For $\mathrm{Bin}(n,p)$ the bound is $np^2 = \lambda p$: the approximation is good when p is small, whatever n is.

Example 8.5 (Birthdays, again)

Let $X_1, X_2, \dots$ be the birthdays of people arriving in sequence and N the number needed for the first shared birthday. For n people, the event $N > n$ means that none of the trials "person j shares with one of persons $1, \dots, j-1$" succeeds; these have success probabilities $(j-1)/365$ given no earlier success, and an exact product gives $P(N > n) = \prod_{j=1}^{n}(1 - \frac{j-1}{365}) \approx \exp\{-\sum_j \frac{j-1}{365}\} = e^{-n(n-1)/730}$, the Poisson-paradigm answer of Example 4.6.

8.6 Selected exercises

Motivation. These are Ross's theoretical exercises for Chapter 8 (book numbering). The first two turn the weak law into a proof of the Weierstrass approximation theorem; the last two give a Markov-type bound for monotone distributions and the optimized Chernoff bound for the binomial.

Exercise 8.4 (Convergence in probability and bounded functions)

Let $Z_1, Z_2, \dots$ be random variables and c a constant such that $P(|Z_n - c| > \epsilon) \to 0$ for every $\epsilon > 0$. Show that $E[g(Z_n)] \to g(c)$ for every bounded continuous g.

Solution.

Step 1. Let $|g| \le M$ and fix $\epsilon > 0$. By continuity at c there is $\delta > 0$ with $|g(x) - g(c)| < \epsilon$ whenever $|x - c| \le \delta$.

Step 2. Split according to whether $|Z_n - c| \le \delta$. On that event $|g(Z_n) - g(c)| < \epsilon$, and on its complement $|g(Z_n) - g(c)| \le 2M$. Hence

$$|E[g(Z_n)] - g(c)| \le E|g(Z_n) - g(c)| \le \epsilon + 2M\, P(|Z_n - c| > \delta).$$

Step 3. The last probability tends to 0, so $\limsup_n |E[g(Z_n)] - g(c)| \le \epsilon$. As ϵ is arbitrary, $E[g(Z_n)] \to g(c)$. ∎

Exercise 8.5 (Bernstein polynomials)

Let f be continuous on $[0,1]$ and

$$B_n(x) = \sum_{k=0}^{n} f\Big(\frac{k}{n}\Big)\binom{n}{k} x^k (1-x)^{n-k}.$$

Prove that $B_n(x) \to f(x)$, uniformly in $x \in [0,1]$. (This is a probabilistic proof of the Weierstrass approximation theorem.)

Solution.

Step 1. Fix $x \in [0,1]$ and let $S_n \sim \mathrm{Bin}(n,x)$, the number of successes in n independent trials with success probability x. Then $B_n(x) = E[f(S_n/n)]$, and $E[S_n/n] = x$, $\mathrm{Var}(S_n/n) = x(1-x)/n \le 1/(4n)$.

Step 2. f is bounded, say $|f| \le M$, and uniformly continuous on $[0,1]$: given $\epsilon > 0$ there is $\delta > 0$ with $|f(u) - f(v)| < \epsilon$ whenever $|u - v| \le \delta$.

Step 3. As in Exercise 8.4 (with $c = x$; note $S_n/n \in [0,1]$, where f is defined) and by Chebyshev's inequality (Proposition 8.2),

$$|B_n(x) - f(x)| \le \epsilon + 2M\,P\Big(\Big|\frac{S_n}{n} - x\Big| > \delta\Big) \le \epsilon + \frac{2M}{4n\delta^2}.$$

Step 4. The right side does not depend on x, so $\limsup_n \sup_{x\in[0,1]} |B_n(x) - f(x)| \le \epsilon$ for every $\epsilon > 0$. Hence $B_n \to f$ uniformly on $[0,1]$, and every continuous function on $[0,1]$ is a uniform limit of polynomials. ■

Exercise 8.6(a) (Monotone probability mass functions)

Let X take values in $\{1, 2, \dots\}$ with $P(X = k)$ nonincreasing in k. Prove that $P(X = k) \le 2E[X]/k^2$.

Solution.

Step 1. Let $p_i = P(X = i)$. Dropping the terms with $i > k$, which are nonnegative,

$$E[X] = \sum_{i\ge 1} i p_i \ge \sum_{i=1}^{k} i p_i.$$

Step 2. Since $p_i \ge p_k$ for $i \le k$, $\sum_{i=1}^{k} i p_i \ge p_k \sum_{i=1}^{k} i = p_k \frac{k(k+1)}{2} \ge p_k \frac{k^2}{2}$. Combining, $p_k \le 2E[X]/k^2$. ■

Exercise 8.11 (The Chernoff bound for the binomial)

Let $X \sim \mathrm{Bin}(n,p)$, $0 < p < 1$, $q = 1 - p$, and $np < i < n$.
(a) Show that $\min_{t>0} e^{-ti} E[e^{tX}]$ is attained where $e^t = iq/\{(n-i)p\}$.
(b) Deduce

$$P(X \ge i) \le \frac{n^n}{i^i (n-i)^{n-i}}\, p^i q^{n-i}.$$

Solution.

(a)

Step 1. $E[e^{tX}] = (pe^t + q)^n$, so we minimize $h(t) = -ti + n\log(pe^t + q)$ over $t > 0$. Here

$$h'(t) = -i + \frac{npe^t}{pe^t + q}, \qquad h''(t) = \frac{npqe^t}{(pe^t + q)^2} > 0,$$

so h is strictly convex and any zero of h' is its global minimum.

Step 2. $h'(t) = 0$ iff $npe^t = i(pe^t + q)$, i.e. $e^t = iq/\{(n-i)p\}$ (which needs $i < n$). This t is positive: $iq > (n-i)p$ is equivalent to $i > np$.

(b) At this t, $pe^t + q = \frac{iq}{n-i} + q = \frac{nq}{n-i}$ and $e^{-ti} = \big(\frac{(n-i)p}{iq}\big)^i$. By Proposition 8.7,

$$P(X \ge i) \le \Big(\frac{(n-i)p}{iq}\Big)^i \Big(\frac{nq}{n-i}\Big)^n = \frac{n^n}{i^i (n-i)^{n-i}}\, p^i q^{n-i}.$$

For instance, with $n = 100$, $p = \frac{1}{2}$ and $i = 60$ the bound is about 0.134; the exact value is about 0.028. ■